From these equations we can calculate the kinetic energy of the ^{234}Th daughter to be 0.072 MeV, while that of the α-particle is 4.202 MeV. Because of the large mass difference between the α-emitting nucleus and the helium atom, almost all of the energy is carried away with the α-particle.

Although the kinetic energy of the daughter nucleus is small in comparison with that of the α-particle, it is large (72 000 eV) in comparison with chemical binding energies (<5 eV). Thus the recoiling daughter easily breaks all chemical bonds by which it is bound to other atoms.

In 1904 it was observed by H. Brooks that measurements on ^{218}Po (RaA), obtained from radon, led to a contamination of the detection chamber by ^{214}Pb (RaB) and ^{214}Bi (RaC). This was explained by Rutherford as being due to daughter recoil in the α-decay of ^{218}Po in the sequence (written symbolically):

$$^{222}\mathrm{Rn}\left(\alpha, 3.8\ \mathrm{d}\right)\,^{218}\mathrm{Po}\left(\alpha, 3.05\ \mathrm{min}\right)\,^{214}\mathrm{Pb}\left(\beta^{-}, 27\ \mathrm{min}\right)\,^{214}\mathrm{Bi}\left(\beta^{-}, 20\ \mathrm{min}\right)\ldots \quad (5.19)$$

This recoil led to ejection of ^{214}Pb into the wall of the instrument. The use of the *recoil* of the daughter to effect its separation was employed by O. Hahn beginning in 1909 and played a central role in elucidating the different natural radioactive decay chains.

The recoil may affect such chemical properties as the solubility or dissolution rate of compounds. For example the dissolution of uranium from uranium rich minerals is considerably higher than one would expect from laboratory solubility data because α and U-atom recoil have moved U-atoms away from their normal sites in the mineral.

Alpha-decay energies are most precisely measured in magnetic spectrometers. From (3.6) and (3.11) it is calculated that

$$E_\alpha = 2\mathbf{e}^2 B^2 r^2 / m_{\mathrm{He}} \quad (5.20)$$

From knowledge of the values of **e**, m_{He}, B, and r, E_α can be calculated. A more common technique is to use *semiconductor detectors* combined with pulse height analyzers ("α-spectrometers", Ch. 9).

5.4. BETA DECAY

5.4.1. Detection

Energetic electrons cause ionization and molecular excitation in matter, although the effect is weaker and more difficult to detect than for α-particles. As a result the effect must be amplified for counting individual β-particles. Ionization is used in *proportional* and *Geiger counters*. Scintillation counting can also be used with various detector systems (§9.5).

5.4.2. The β-decay process

The radioactive decay processes which are designated by the general name of β-decay include electron emission (β^- or $^0_{-1}e$), positron emission (β^+ or $^0_{+1}e$) and electron capture (EC). If we use the β-decay of ^{137}Cs as an example, we can write

$$^{137}\text{Cs} \rightarrow {}^{137\text{m}}\text{Ba} + \beta^- \tag{5.21}$$

This β-decay must occur between discrete quantum levels of the parent nuclide ^{137}Cs and the daughter nuclide $^{137\text{m}}$Ba.

The quantum levels of nuclei are characterized by several quantum numbers, an important one being the *nuclear spin*. The spin value for the ^{137}Cs ground state level is 7/2, while that of $^{137\text{m}}$Ba is 11/2. The electron emitted is an elementary particle of spin 1/2. In nuclear reactions the nuclear angular momentum must be conserved (5.10), which means that in radioactive decay processes the difference in total spin between reactant and products must be an integral value (5.12). Inspection of our example shows that this conservation of spin rule is violated if the reaction is complete as we have written it. The sum of the spin of the $^{137\text{m}}$Ba and of the electron is 11/2 + 1/2 or 6, while that of the ^{137}Cs is 7/2. Therefore, the change in spin (ΔI) in the process would seem to be 5/2 spin units. In as much as this is a non-integral value, it violates the rule for conservation of angular momentum. Before accounting for this discrepancy let us consider another aspect of β-decay which seems unusual. Figure 5.1 shows the β-particle spectrum of

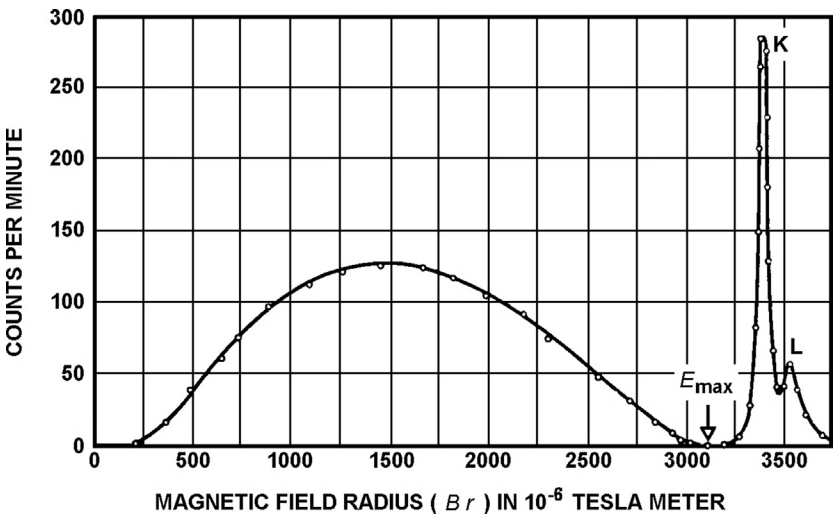


Figure 5.1 Spectrum for electrons emitted by ^{137}Cs as observed with a magnetic spectrometer. $Br \propto \sqrt{E}$, cf. Eqn (5.22). *(From W. Gentner, H. Maier-Leibnitz, and W. Bothe.)*

^{137}Cs as obtained by a magnetic spectrometer. The β-particle energy is calculated by the relation

$$E_\beta = \mathbf{e}^2 B^2 r^2/(2\, m_e) \tag{5.22}$$

where m_e is the electron relativistic mass, cf (2.9). The spectrum shows the number of β-particles as a function of *Br*, which is proportional to $\sqrt{E_\beta}$ through (5.22). We observe a continuous distribution of energies. This seems to disagree with our earlier statement that decay occurs by change of one nucleus in a definite energy state to another nucleus also in a definite energy state. The two sharp peaks designated K and L at the high energy end of the spectrum are not related to the beta spectrum itself and are discussed later in the chapter (§5.5.2).

5.4.3. The neutrino

This problem of "wrong" spin change and the continuous "non-quantized" spectrum led W. Pauli to the assumption that β-decay involves emission of still another particle which has been named the *neutrino* and given the symbol ν, cf. §2.5. The neutrino has a spin value of 1/2, an electric charge of 0, and a mass ≈ 0. It is therefore somewhat similar to the photon, which has neither mass, electric charge nor spin. However, while the photon readily interacts with matter, the neutrino does not. In fact the interaction is so unlikely that a neutrino has a very high probability of passing through the entire earth without reacting.

The spin attributed to the neutrino allows conservation of angular momentum; in our example, the total spin of the products would be $11/2 + 1/2 + 1/2$ or 13/2, and when the spin of ^{137}Cs, 7/2, is subtracted from this the result is $6/2 = 3$ which is an acceptable integral value. Thus the decay reaction above is incomplete and must be written

$$^{137}\mathrm{Cs} \;\rightarrow\; ^{137\mathrm{m}}\mathrm{Ba} + \beta^- + \overline{\nu} \tag{5.23}$$

Notice we have replaced ν by $\overline{\nu}$, which is the designation of the *antineutrino*. Anti-neutrinos are emitted in electron decay while "regular" neutrinos ν in positron decay. We can here consider the particles identical; cf. §2.3. Because of the extremely low probability of interaction or neutrinos with matter, they are unfortunately often omitted in writing β-decay reactions.

The neutrino theory also explains the energy spectrum in β-decay. However, this necessitates the introduction of another important nuclear concept, that of *relativistic mass* and *rest mass*. In 1901 S. G. Kaufmann showed in experiments that the mass of an electron *m* seemed to increase when its velocity *v* approached that of the speed of light **c**. Later, it was found that this increase followed an expression

$$m = m^o\left(1 - v^2/\mathbf{c}^2\right)^{-1/2} \tag{5.24}$$

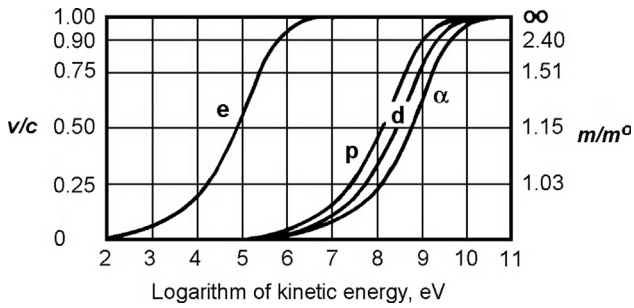


Figure 5.2 Relativistic masses m for some common nuclear particles, divided by their rest masses m^0, as a function of the kinetic energy of the particle.

based on H. Lorentz's studies of the relation between distance, speed of light, and time. m^o is the rest mass of the particle (at velocity $v = 0$), while m is referred to as the relativistic mass. This relation is valid for any moving object, macroscopic or microscopic, whether it is a "particle", a "wave-packet" or a space rocket. Figure 5.2 shows the quotient $v/\mathbf{c}$ as a function of the kinetic energy of the particle, E_{kin} (in some literature the quotient $v/\mathbf{c}$ is designated as β but for clarity compared to the radioactive decay we have not used β).

If the parentheses in (5.24) is expanded by means of the binomial theorem of algebra, it approximates to

$$m = m^o + \tfrac{1}{2}\, m^o v^2/\mathbf{c}^2 \tag{5.25}$$

The last term is approximately the kinetic energy of the particle (3.6) divided by $\mathbf{c}^2$, and thus

$$m \approx m^o + E_{\text{kin}}/\mathbf{c}^2 \tag{5.26}$$

The increase in mass, $\Delta m = m - m^o$, is due to the kinetic energy of the particle,

$$E_{\text{kin}} = \Delta m \mathbf{c}^2 \tag{5.27}$$

This was generalized by A. Einstein in the special theory of relativity, leading (after more detailed calculations) to the well known mass-energy relationship

$$E = m\mathbf{c}^2 \tag{5.28}$$

which we already have applied in the discussion of the nuclear binding energy (4.3).

When a neutrino is ejected from the nucleus it carries away energy of kinetic nature. Thus, according to (5.26) the neutrino has a relativistic mass > 0, and obviously also a momentum $p = mv$. Recoil studies of β-decay have proven this to be true.

In order to correctly apply (5.22) for the calculation of the β-decay energy, the relativistic electron mass must be used; as is seen from Figure 5.2, already at 0.1 MeV, the relativistic mass of the electron is 15% larger than the rest mass $\mathbf{m}_e^o$. (In the following

the rest masses of the electron, neutron, etc., will be denoted simply as $\mathbf{m}_e$, $\mathbf{m}_n$, etc.; capital **M** if in universal mass units, u)

The energy released in β-decay is distributed between the neutrino, the electron, and the recoil of the daughter nucleus. This latter will be much smaller than the first two and can be neglected in a first approximation (§5.4.8). Therefore, the total β-decay energy can be considered to be distributed between the neutrino and the electron. For the decay $^{137}Cs \rightarrow {}^{137m}Ba$ it can be shown that the total decay energy Q_β is 0.514 MeV. This is also termed E_{max}. The neutrino energy spectrum is the complement of the β-particle energy spectrum − summing up to E_{max}. If the energy of the electron is 0.400 MeV, that of the neutrino is 0.114 MeV. If the electron energy is 0.114 MeV, the neutrino energy is 0.400 MeV.

In β^--decay the average value of the β^--particle energy is about 0.3 E_{max} due to attraction by the nuclear charge, and approximately given by the empirical equation (5.29).

$$E_{average} \approx 0.33 E_{max} \left(1 - \frac{\sqrt{Z}}{50}\right)\left(1 + \frac{\sqrt{E_{max}}}{4}\right) \quad \text{for } \beta^- \tag{5.29}$$

In positron emission, the average energy of the β^+-particle is higher and approximately 0.4 E_{max} as shown by the empirical equation (5.30).

$$E_{average} \approx 0.33 E_{max} \left(1 + \frac{\sqrt{E_{max}}}{4}\right) \quad \text{for } \beta^+ \tag{5.30}$$

With a few exceptions, equations (5.29) and (5.30) reproduce measured $E_{average}$ values to within 5%.

The assumption that the neutrino has a zero rest mass has been questioned by experimentalists and theorists. A number of experiments at the Kamiokande facility have established an upper limit of the rest mass of the electron neutrino as ~0.07 eV. The implications of a finite rest mass are broad as the nature of the neutrino and the theory of beta decay is involved.

5.4.4. Double beta decay

The rather unusual (and very slow) $\beta^-\beta^-$-decay mode is energetically possible for several even-even nuclei, and kinetically possible to observe in those cases where the separating odd-odd nucleus of higher energy prevents normal β^--decay, see Fig. 4.6 V. It has recently been observed for some such cases, e.g.

$$^{82}_{34}Se \rightarrow {}^{82}_{36}Kr + 2\beta^- + 2\overline{\nu} \tag{5.31}$$

The half-life of about 1.7×10^{20} y can be observed because small amounts of the noble gas Kr can be physically isolated from large amounts of ^{82}Se (9% natural

abundance), and then measured. Studies of ββ-decay are of importance for evaluation of neutrino properties. However, neutrinoless ββ-decay may have been observed in the decay of $^{76}_{32}$Ge.

5.4.5. β^--Decay

This process can be written symbolically as follows:

$$^{A}_{Z}\mathrm{X} \rightarrow {}^{A}_{Z+1}\mathrm{X} + {}^{0}_{-1}\beta + \overline{\nu} \tag{5.32}$$

However, if we take the electrons into account, the neutral parent atom has Z orbital electrons, while the daughter atom, with a nuclear charge of $Z + 1$, must capture an electron from the surroundings, in order to become neutral:

$$^{A}_{Z+1}\mathrm{X}^{+} + \mathrm{e}^{-} \rightarrow {}^{A}_{Z+1}\mathrm{X} \tag{5.33}$$

Moreover, since the beta particle emitted provides an electron to the surroundings, the total electron balance remains constant. As a result, in the calculation of the decay energy it is not necessary to include the mass of the emitted β-particle as the use of the mass of the neutral daughter atom includes the extra electron mass. The equation for calculating the Q-value β^- decay is thus:

$$Q_{\beta^-} = -931.5\,(M_{Z+1} - M_Z) \tag{5.34}$$

As an example we can take the decay of a free neutron in a vacuum; it transforms spontaneously with a half-life of 10.6 min to a proton through beta decay.

$$^{1}_{0}\mathrm{n} \rightarrow {}^{1}_{1}\mathrm{H} + {}^{0}_{-1}\mathrm{e}^{-} \tag{5.35}$$

The Q-value for this reaction is

$$Q_{\beta^-} = -931.5\,(1.007\,825 - 1.008\,665) = 0.782\ \mathrm{MeV} \tag{5.36}$$

5.4.6. Positron decay

Positron decay can be written symbolically as

$$^{A}_{Z}\mathrm{X} \rightarrow {}^{A}_{Z-1}\mathrm{X}^{-} + {}^{0}_{+1}\beta + \nu \rightarrow {}^{A}_{Z-1}\mathrm{X} + {}^{0}_{-1}\mathrm{e}^{-} + {}^{0}_{+1}\beta + \nu \tag{5.37}$$

Here we must consider the net atomic charges. The daughter nucleus has an atomic number one less than the parent. This means that there will be one electron mass associated with the change in atomic number. Moreover, an electron mass must also be included for the positive electron emitted. When $^{22}_{11}$Na decays to $^{22}_{10}$Ne, there are 11 electrons included in the ^{22}Na atomic mass but only 10 in the ^{22}Ne atomic mass. Consequently, an extra electron mass must be added on the product side in addition to

the electron mass associated with the positron particle. The calculation of the Q-value must therefore include two electron masses beyond that of the neutral atoms of the parent and daughter

$$Q_{\beta+} = -931.5\left(M^{A}_{Z-1} + 2\,\mathbf{M}_{e} - M^{A}_{Z}\right) \tag{5.38}$$

Each electron mass has an energy equivalent to 0.511 MeV, since $931.5 \times 0.000\,549 = 0.511$.

Consider the calculation of the Q-value for the reaction

$$^{13}_{7}\mathrm{N} \rightarrow {}^{13}_{6}\mathrm{C} + \beta^{+} \tag{5.39}$$

For this reaction we have

$$Q_{\beta+} = -931.5\left(13.003\,355 - 13.005\,739\right) - 2 \times 0.511 = 1.20\ \mathrm{MeV} \tag{5.40}$$

5.4.7. Electron capture

The EC decay process can be written symbolically

$$^{A}_{Z}\mathrm{X} \overset{\mathrm{EC}}{\rightarrow} {}^{A}_{Z-1}\mathrm{X} + \nu \tag{5.41}$$

The captured electron comes from one of the inner orbitals of the atom. Depending on the electron shell from which the electron originates, the process is sometimes referred to as K-capture, L-capture, etc. The probability for the capture of an electron from the K-shell is several times greater than that for the capture of an electron from the L-shell, since the wave function of K-electrons is substantially larger at the nucleus than that of L-electrons. Similarly, the probability of capture of electrons in higher order shells decreases with the quantum number of the electron shell.

The calculation of the decay energy in electron capture follows the equation

$$Q_{\mathrm{EC}} = -931.5\left(M^{A}_{Z-1} - M^{A}_{Z}\right) \tag{5.42}$$

Note that like the case of the β^{-} decay, it is not necessary to add or subtract electron masses in the calculation of the Q-value in EC. An example of EC is the decay of ^{7}Be to ^{7}Li for which it is possible to calculate that the Q-value is 0.861 MeV. This reaction is somewhat exceptional since for neutron deficient nuclei with values of Z below 30, positron emission is the normal mode of decay. Electron capture is the predominant mode of decay for neutron deficient nuclei whose atomic number is greater than 80. The two processes compete to differing degrees for the nuclei between atomic numbers 30 and 80. Electron capture is observed through the emission of electrons from secondary reactions occurring in the electron shell because of the elemental change (see §5.9).

5.4.8. Daughter recoil

If the β-particle and the neutrino are emitted with the same momentum but in opposite direction, the daughter nucleus experiences no recoil. On the other hand, if they are both emitted in the same direction, or if all the energy is carried away with one of the particles, the daughter experiences maximum recoil. The daughter therefore recoils with kinetic energies from zero up to a maximum value (when the β-particle is emitted with maximum energy). We can therefore write

$$Q_\beta = E_d + E_{max} \tag{5.43}$$

where E_d is the recoil energy of the daughter nucleus. From the laws of conservation of energy and momentum, and taking the relativistic mass changes of the electron into account, one finds that the daughter recoil energy is

$$E_d = \mathbf{m}_e E_{max}/m_d + E_{max}^2/\left(2\, m_d\, \mathbf{c}^2\right) \tag{5.44}$$

The recoil energy is usually ~100 eV, which still is sufficient for causing atomic rearrangements in surrounding molecules. In the decay of ^{14}C (to ^{14}N), E_{max} is 0.155 MeV, which gives $E_d = 7$ eV. However, by labelling ethane, $^{14}CH_3\,{}^{14}CH_3$, with ^{14}C in both C positions, it was found that $^{14}CH_3NH_2$ was formed in 50% of the cases when one of the ^{14}C atoms in ethane had decayed, although the C≡N bond strength is only 2.1 eV; most of the decays occur with less than the maximum recoil energy, which also can be averaged over the whole molecule. The small recoil also explains why decay reactions like

$$^{127}TeO_3{}^{2-} \rightarrow {}^{127}IO_3{}^{-} + \beta^- \tag{5.45}$$

and

$$^{52}MnO_4{}^{-} \rightarrow {}^{52}CrO_4{}^{2-} + \beta^+ \tag{5.46}$$

are possible, even when E_d is tens of electron volts. However, secondary effects tend to cause the chemical bond to break following radioactive decay (see §5.9).

5.5. GAMMA EMISSION AND INTERNAL CONVERSION

The α- and β-decay may leave the daughter nucleus in an excited state. This excitation energy is removed either by γ-ray emission or by a process called *internal conversion*.

5.5.1. Gamma emission

The α-emission spectrum of ^{212}Bi is shown in Figure 5.3. It is seen that the majority of the α-particles have an energy of 6.04 MeV, but a considerable fraction (~30%) of the α-particles have higher or lower energies. This can be understood if we assume that the